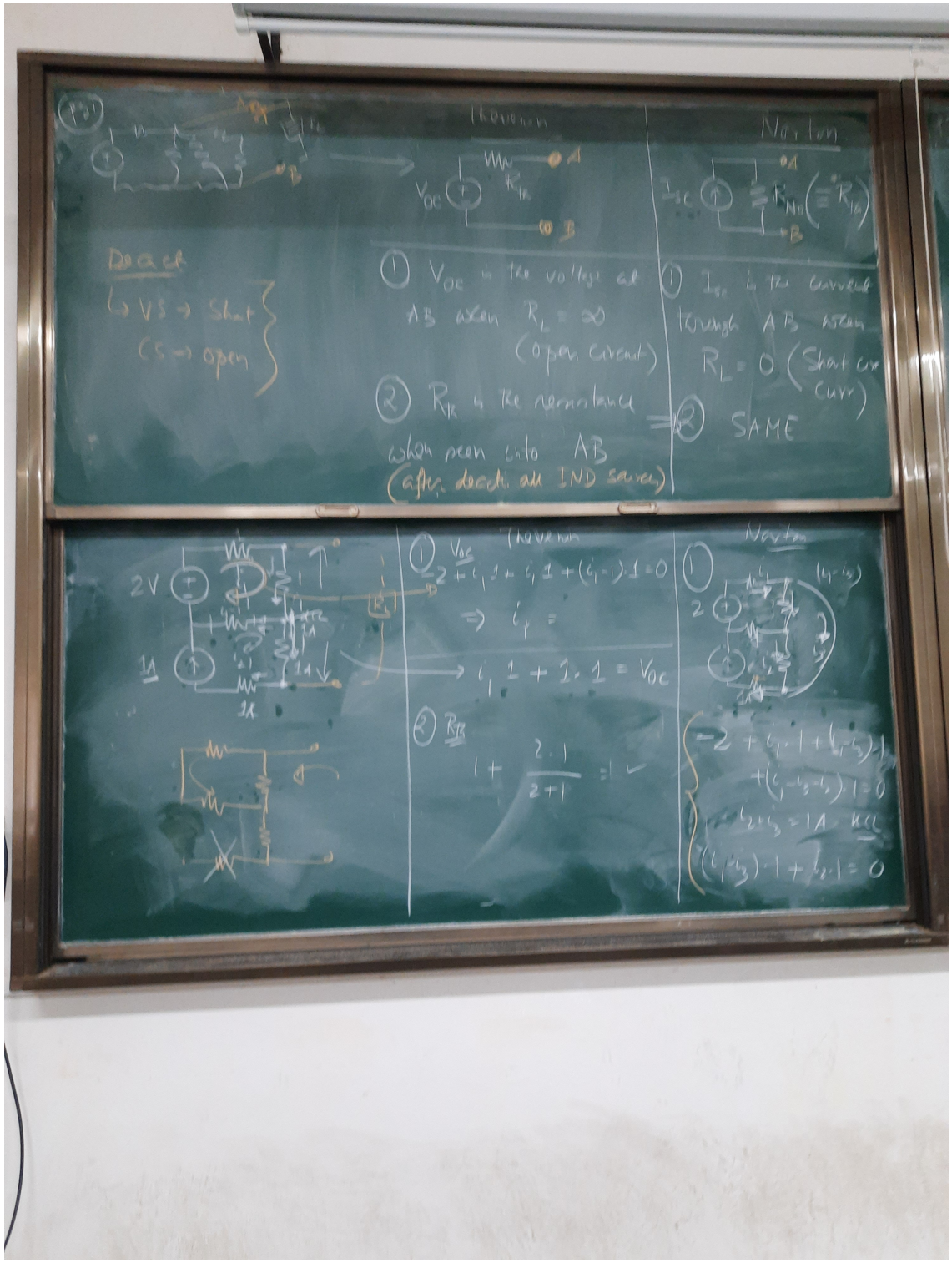
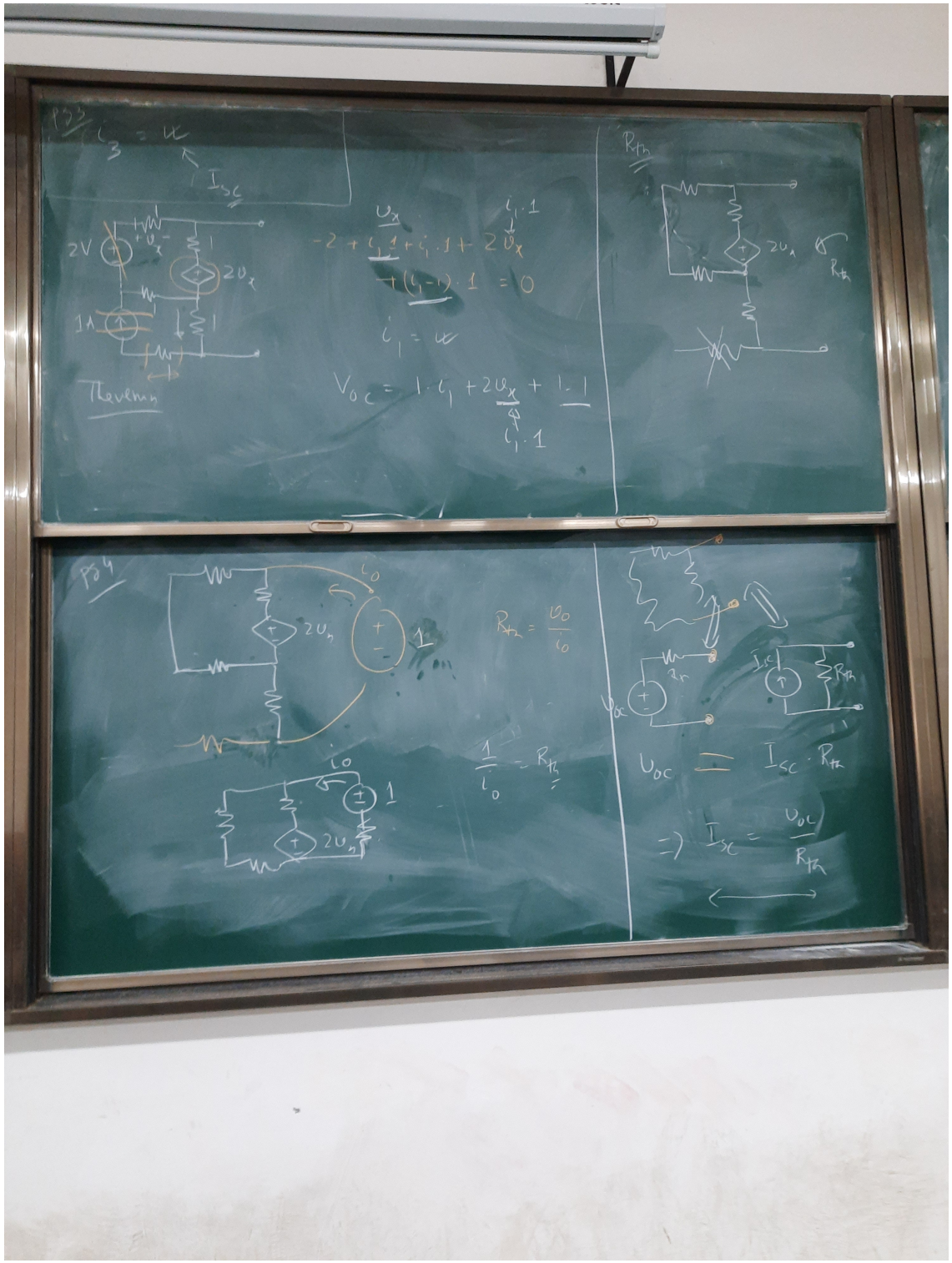
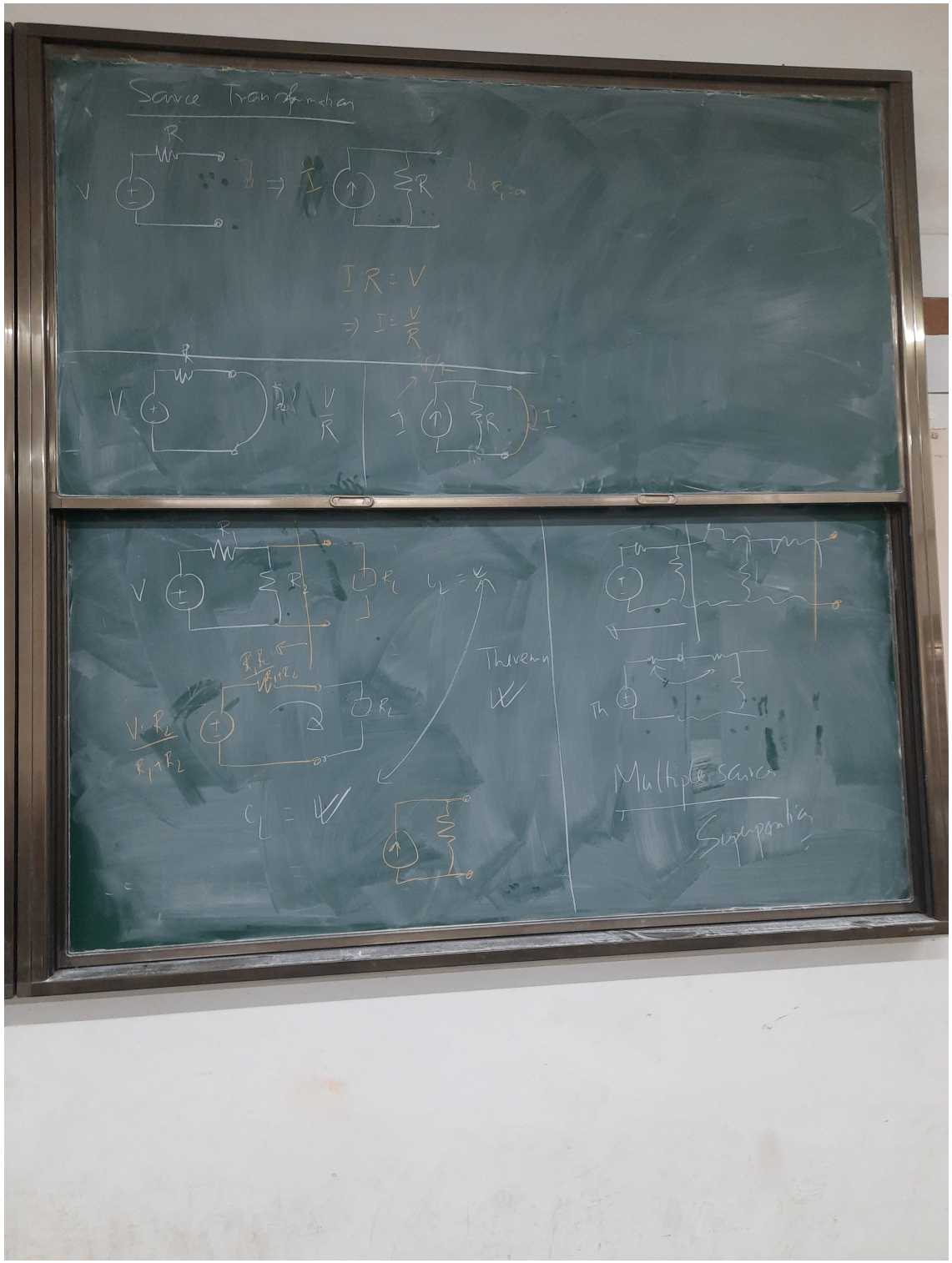

Deact
↳ VS → Short
CS → open
Thevenin
① V_{oc} is the voltage at AB when $R_L = \infty$ (open circuit)
② R_{th} is the resistance when seen into AB (after deact. all IND sources)
Norton
① I_{sc} is the current through AB when $R_L = 0$ (Short ckt curr)
② SAME
2V
1A
1Ω
$i_1 1 + 1 \cdot 1 = V_{oc}$
$1 + \frac{2 \cdot 1}{2+1} = 1$
$-2 + i_1 \cdot 1 + (i_1 - i_3) \cdot 1 + (i_1 - i_3 - i_2) \cdot 1 = 0$
$i_2 + i_3 = 1A$ KCL
$(i_1 - i_3) \cdot 1 + i_2 \cdot 1 = 0$


$i_3 = ?$
I_{sc}
2V
$+ v_x -$
$2v_x$
1A
Thevenin
v_x
$i_1 \cdot 1$
$-2 + i_1 \cdot 1 + i_1 \cdot 1 + 2v_x + (i_1 - 1) \cdot 1 = 0$
$i_1 = ?$
$V_{oc} = 1 \cdot i_1 + 2v_x + 1 \cdot 1$
$i_1 \cdot 1$
R_{th}
$2v_1$
R_{th}
Pg 4
i_o
$2v_n$
1
$R_{th} = \frac{v_o}{i_o}$
i_o
$2v_n$
1
$\frac{1}{i_o} = R_{th}$
I_{sc}
R_{th}
V_{oc}
$V_{oc} = I_{sc} \cdot R_{th}$
$\Rightarrow I_{sc} = \frac{V_{oc}}{R_{th}}$


Source Transformation
R
V
I
R
$R_L = \infty$
$IR = V$
$\Rightarrow I = \frac{V}{R}$
$\frac{V}{R}$
R_1
R_2
R_L
$V_L = ?$
$\frac{V \cdot R_2}{R_1 + R_2}$
$\frac{R_1 R_2}{R_1 + R_2}$
Thevenin
$V_L = ?$
Th
Multiple source
Superposition